

# Computer Vision

## Object Recognition

Soo Min Kimm

21/10/25

- 1 Computer Vision (CV)
- 2 Backbone / Feature Extractor
  - Convolutional Neural Network (CNN)
  - Vision Transformer (ViT)
- 3 Detector
  - R-CNN
  - Fast R-CNN
  - Faster R-CNN
  - YOLO
  - SSD
  - DETR
  - Deformable DETR
  - DINO
- 4 Summary

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# Object Recognition

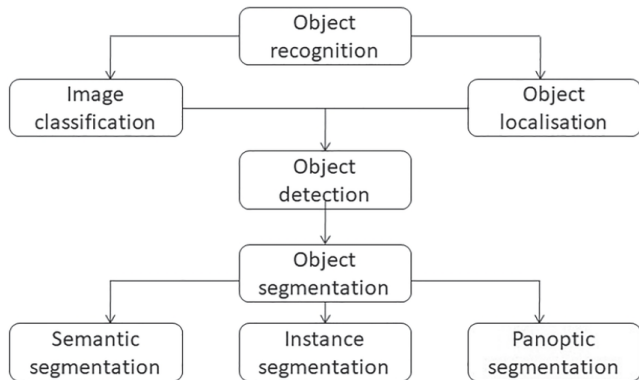


Figure: Overview of object recognition. Image from Brahmi & Jdey (2023) - Automatic tooth instance segmentation and identification from panoramic X-Ray images using deep CNN.

# Example

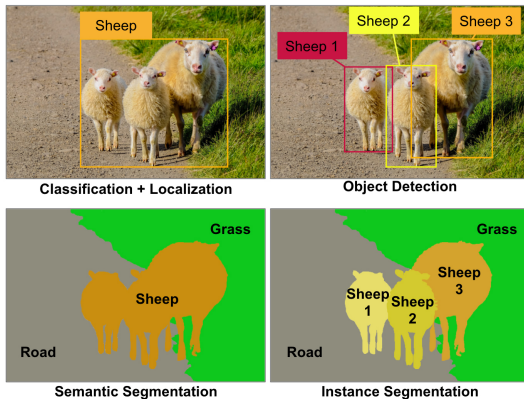


Figure: Example of object recognition. Image from [https://www.oreilly.com/content/introducing-capsule-networks/?cmp=tw-data-na-article-ainy18\\_thea](https://www.oreilly.com/content/introducing-capsule-networks/?cmp=tw-data-na-article-ainy18_thea)

# Pipeline

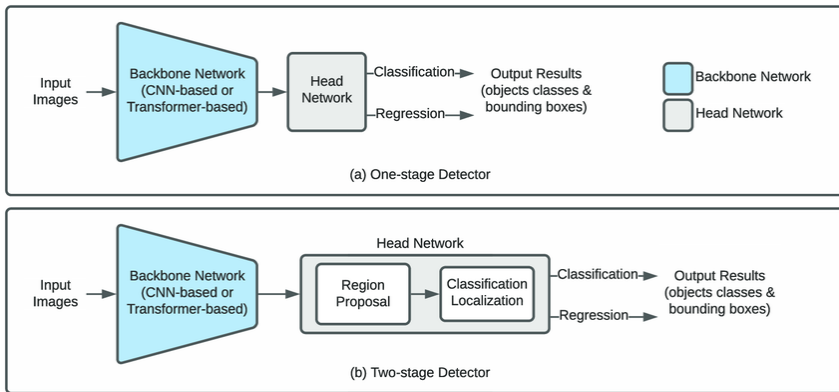


Figure: Pipeline of object detection. Image from Kang et al. (2022) - A Survey of Deep Learning-Based Object Detection Methods and Datasets for Overhead Imagery.

# Evaluation Metrics

## Intersection over Union (IoU)

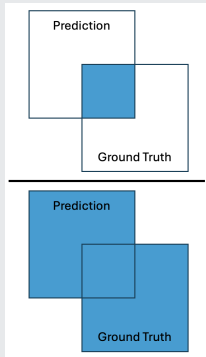


Figure:  $\text{IoU} = \frac{X \cap Y}{X \cup Y}$ .

## Mean Average Precision (mAP)

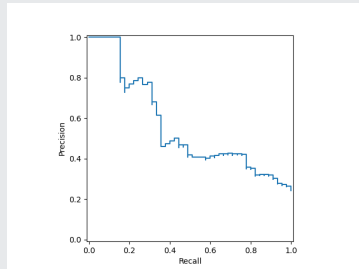


Figure:  $\text{mAP} = \frac{1}{N} \sum_{k=1}^N \text{AP}_i$ .

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# Inductive Bias

A learning algorithm seeks a hypothesis  $h \in \mathcal{H}$  that maps input  $x$  to outputs  $y$ . Given training examples  $D = \{(x_i, y_i)\}$ , there are typically infinitely many  $h$  that can fit  $D$ .

| Type                           | Description                                                       | Example in Models                                                    |
|--------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Architectural Bias</b>      | Structure or connectivity built into the model.                   | CNN's local kernels, RNN's recurrence, Transformer's self-attention. |
| <b>Regularization Bias</b>     | Added terms in the loss that prefer certain solutions.            | Weight decay, dropout, early stopping.                               |
| <b>Optimization Bias</b>       | The way the model is trained can favor particular minima.         | SGD tends to find flatter minima (better generalization).            |
| <b>Data Bias</b>               | Augmentations or preprocessing that shape how data are "seen."    | Rotations, flips, normalization.                                     |
| <b>Prior Distribution Bias</b> | In Bayesian learning, priors encode assumptions about parameters. | Gaussian prior in Bayesian linear regression.                        |

Table: Types of inductive bias.

# Receptive Field

For a unit at layer  $L$  and spatial position  $(i, j)$ , the (theoretical) receptive field is the set of input pixels whose values can possibly affect that unit, given the network's filters, strides, dilations, and paddings.

- Theoretical RF: The Maximal geometric region implied by the architecture.
- Effective RF: The actual influence distribution within that region when the networks is trained—typically strongest at the centre and decaying outward.

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# Architecture

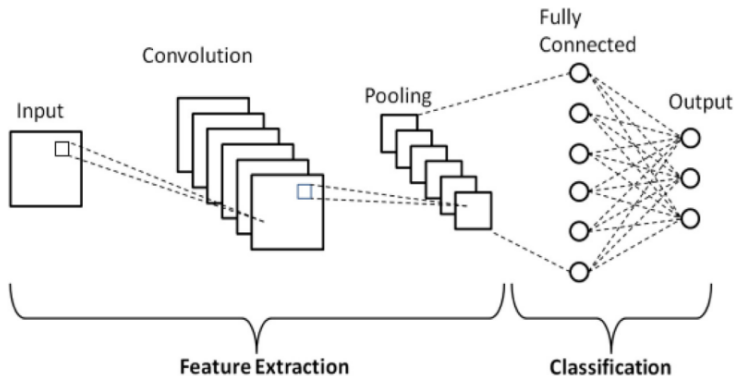


Figure: CNN. Image from Phung & Rhee (2018) - A Deep Learning Approach for Classification of Cloud Image Patches on Small Datasets.

# Definition

## Kernel

Small learnable filter that extracts local patterns.

## Padding

Adding zeros around the input to preserve spatial dimensions.

## Dilation

Spacing elements in the kernel to enlarge receptive fields w/o more parameters.

## Stride

Step size when sliding kernel that controls downsampling.

## Feature (Activation) Map

The output tensor of a layer showing where and how strongly learned patterns appear.

Early feature maps: Edges, blobs, textures.

Deep feature maps: Object parts, semantics.

# Convolution

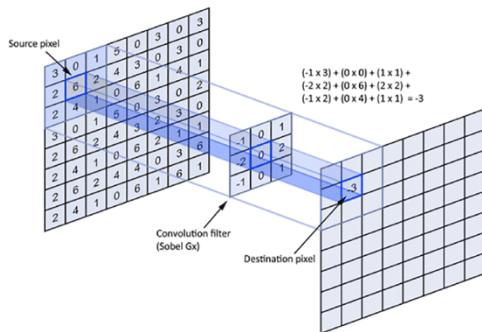


Figure: Convolution. Image from Almryad & Kutuku (2020) - Automatic Identification for Field Butterflies by CNN

# Pooling

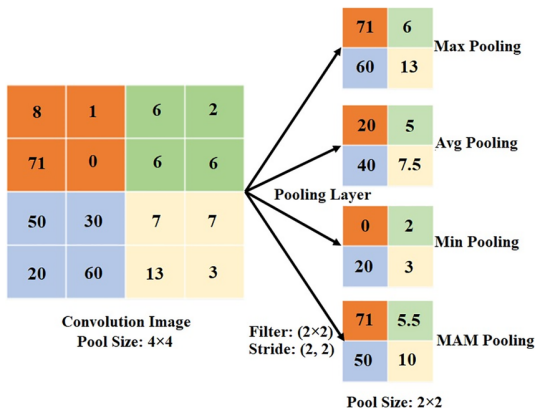


Figure: Pooling. Image from Akgül (2024) - A Pooling Method Developed for Use in Convolutional Neural Networks.

# VGG: Architecture

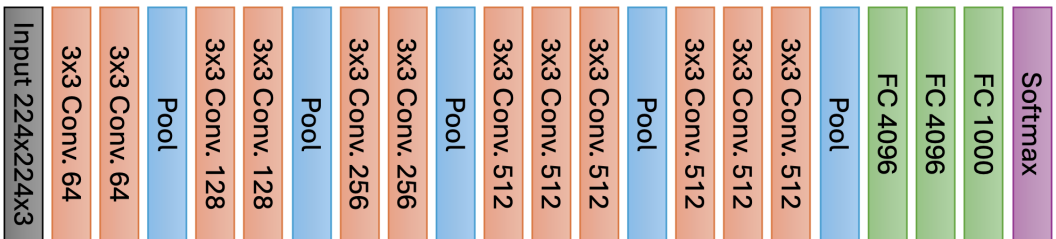


Figure: VGG.

# Inception: Architecture

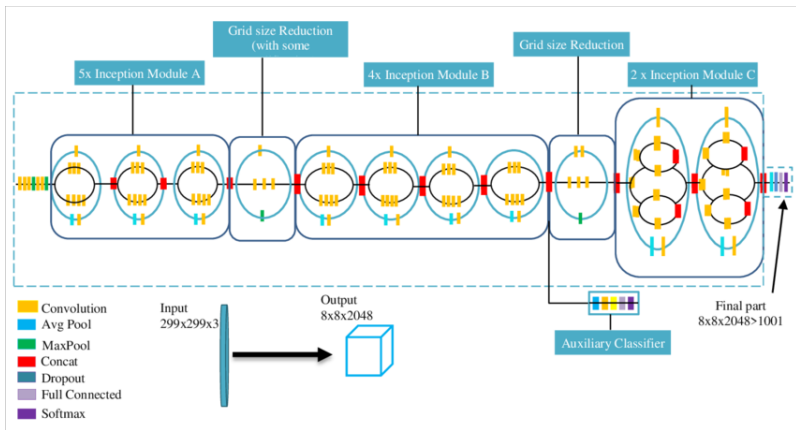


Figure: Inception. Image from Iparraguirre-Villanueva (2022) - Convolutional Neural Networks with Transfer Learning for Pneumonia Detection.

# VGG: $1 \times 1$ Convolution

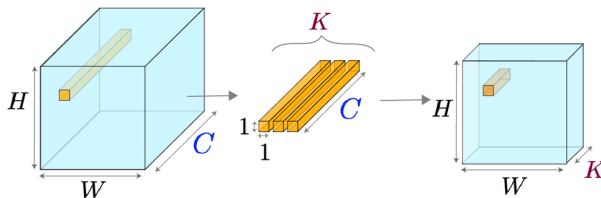


Figure:  $1 \times 1$  convolution. Image from <https://cvml-expertguide.net/terms/dl/layers/convolution-layer/1x1-convolution/>.

- Combine information from different channels at the same spatial location
- Reduce channel dimension

# ResNet: Architecture

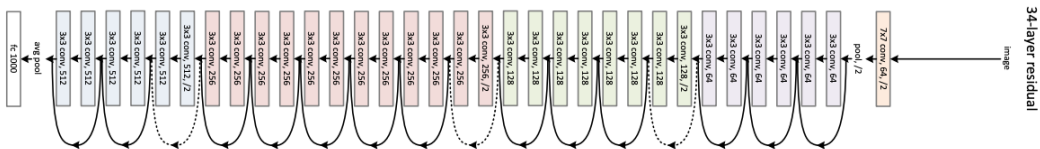


Figure: ResNet. Image from He et al. (2015) - Deep Residual Learning for Image Recognition.

# ResNet: Residual Connection

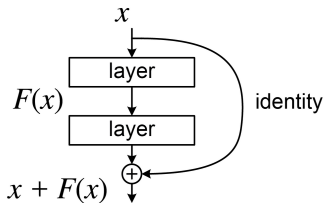


Figure: Residual connection. Image from [https://www.wikipedia.com/en/articles/Residual\\_neural\\_network](https://www.wikipedia.com/en/articles/Residual_neural_network).

- Stable training by preserving gradient flow

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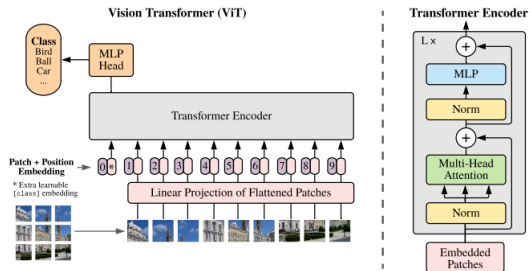


Figure: ViT. Image from Dosovitskiy et al. (2020) - An Image is Worth 16x16 Words.

- Class token summarises the entire sequence

# DeiT: Architecture

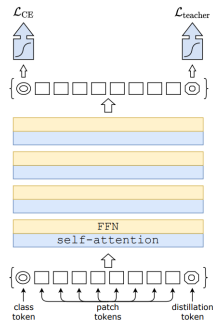


Figure: DeiT. Image from Touvron et al. (2020) - Training Data-Efficient Image Transformers & Distillation through Attention.

- Class token learns from ground truth labels
- Distillation token learns from teacher's outputs

# DeiT: Knowledge Distillation

DeiT is trained as following:

- 1 Train the CNN teacher and freeze the parameters once trained.
- 2 Train the ViT student to learn from the ground truth and the teacher's predictions.

Loss is given as following:

- Soft distillation:  $\mathcal{L} = (1 - \lambda)\mathcal{L}_{\text{CE}}(\psi(Z_s), y) + \lambda\tau^2\mathcal{L}_{\text{KL}}(\psi(\frac{Z_s}{\tau}), \psi(\frac{Z_t}{\tau}))$
- Hard distillation:  $\mathcal{L} = \frac{1}{2}\mathcal{L}_{\text{CE}}(\psi(Z_s), y) + \frac{1}{2}\mathcal{L}_{\text{CE}}(\psi(Z_t), y_t)$

where  $Z_t, Z_s$  are logits of teacher and student models respectively,  $\tau$  is the distillation temperature,  $\lambda$  is the coefficient balancing  $\mathcal{L}_{\text{KL}}$ ,  $y_t = \arg \max_c Z_t(c)$ , and  $\psi$  is the softmax function.

# Swin Transformer: Architecture

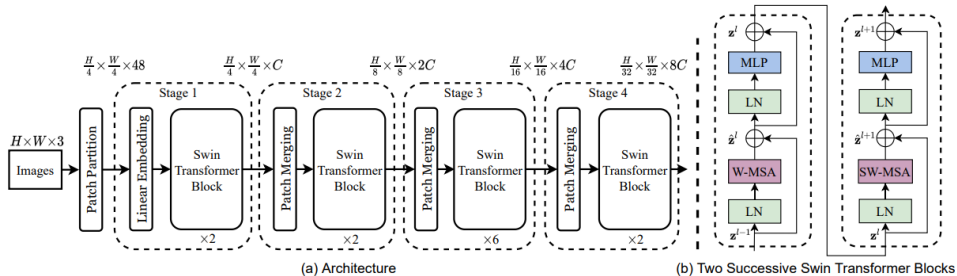


Figure: Swin Transformer. Image from Liu et al. (2021) - Swin Transformer.

# Swin Transformer: Shifted Window

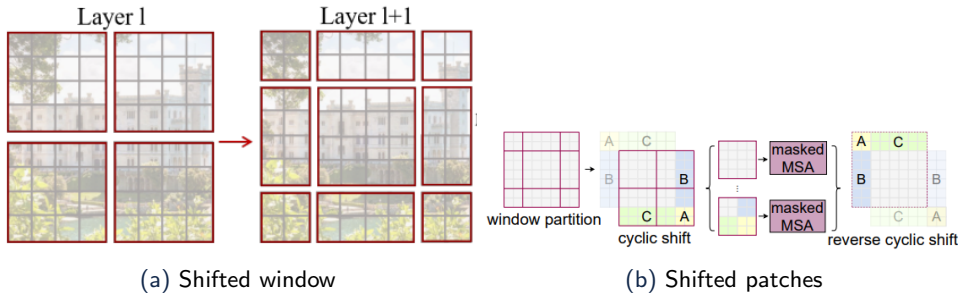


Figure: Shifted window partitioning. Image from Liu et al. (2021) - Swin Transformer.

# Swin Transformer: Hierarchy

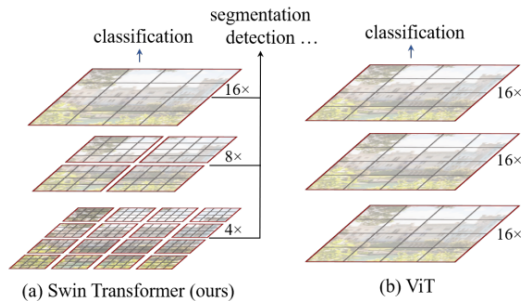


Figure: Hierarchy of Swin Transformer. Image from Liu et al. (2021) - Swin Transformer.

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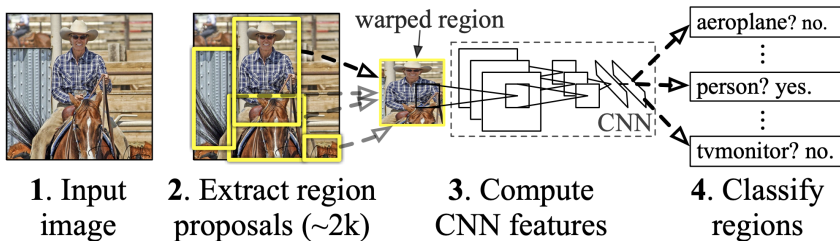


Figure: R-CNN. Image from Girshick et al. (2014) - Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation.

# Selective Search

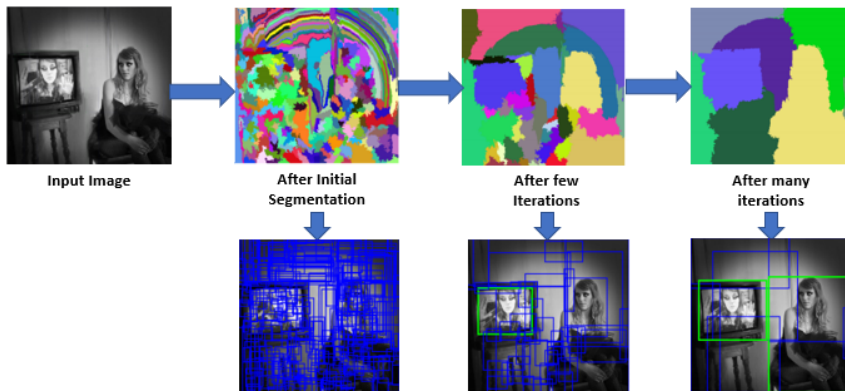


Figure: Selective search algorithm. Image from Uijlings et al. (2013) - Selective Search for Object Recognition.

# Hard Negative Mining

# Non-Maximum Suppression

# Contributions & Limitations

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# RoI Pooling

# Truncated SVD

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# Region Proposal Network (RPN)

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# Hungarian Matching Algorithm

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# Deformable Convolution

# Deformable Attention

# Contributions & Limitations

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# Self-Supervised Knowledge Distillation

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# Summary 1

| Model           | Stage     | Proposals / Anchors                 | NMS | Matching              |
|-----------------|-----------|-------------------------------------|-----|-----------------------|
| R-CNN           | 2-stage   | Selective Search (2k)               | Yes | IoU-based             |
| Fast R-CNN      | 2-stage   | Selective Search (external)         | Yes | IoU-based             |
| Faster R-CNN    | 2-stage   | RPN (anchor-based)                  | Yes | IoU / anchor assign   |
| YOLO (v1)       | 1-stage   | Grid-based boxes (anchor-free-ish)  | Yes | Grid responsibility   |
| SSD             | 1-stage   | Default boxes (multi-scale anchors) | Yes | IoU / anchor assign   |
| DETR            | Set-based | None (no anchors)                   | No  | Hungarian (bipartite) |
| Deformable DETR | Set-based | None                                | No  | Hungarian (bipartite) |
| DINO            | Set-based | None                                | No  | Hungarian (bipartite) |

Table: Properties of detectors.

# Summary 2

| Model           | Key Ideas / Strengths                                                  | Limitations / Notes                                                    |
|-----------------|------------------------------------------------------------------------|------------------------------------------------------------------------|
| R-CNN           | Deep features for detection; strong accuracy for its time.             | Very slow (per-RoI CNN); no shared conv features; multi-step pipeline. |
| Fast R-CNN      | Single CNN for image; RoI Pooling; joint classification & regression.  | Still depends on external proposals; not real-time.                    |
| Faster R-CNN    | Region Proposal Network (RPN); shared conv features; end-to-end.       | Requires anchor tuning; slower than one-stage.                         |
| YOLO (v1)       | Real-time, single forward pass; simple end-to-end training.            | Struggles with small/overlapping objects; coarse grid.                 |
| SSD             | Multi-scale feature maps; fast and lightweight.                        | Fixed anchors; limited context; requires tuning.                       |
| DETR            | End-to-end set prediction; no anchors/NMS; simple loss.                | Slow convergence; weak for small objects.                              |
| Deformable DETR | Deformable multi-scale attention; faster convergence; sparse sampling. | Complex implementation; heavy computation.                             |
| DINO            | Self-distillation; denoising queries; strong accuracy.                 | Transformer latency; careful training recipe needed.                   |

Table: Overview of detectors.